



Is it possible to make Decompose work with coefficients containing radicals?

Asked 6 years, 11 months ago Modified 6 years, 11 months ago Viewed 143 times



5

It appears that [Decompose](#) works reliably only on polynomials with integer coefficients, although it seems to be not mentioned in the docs. Can I make it work (or implement an alternative that would work) on polynomials with coefficients containing radicals? For example,



```
Decompose[3 + 3 √2 + (14 + 4 √2) x + (12 + 26 √2) x^2 +
(56 + 8 √2) x^3 + (8 + 48 √2) x^4 + 48 x^5 + 16 √2 x^6, x]
```



```
(* {3 + 3 √2 + (14 + 4 √2) x + (4 + 12 √2) x^2 + 8 x^3, x + √2 x^2} *)
```



Is there a known algorithmic solution to this problem at all?

programming

symbolic

polynomials

algebraic-manipulation

Share Edit Close Delete Flag

asked Sep 21, 2019 at 22:49



Vladimir Reshetnikov

6,651 32 83

1 Answer

Sorted by: [Reset to default](#) Date modified (newest first) ▾



9

One way to decompose a polynomial $p(x)$ is to factor $p'(x)$ over a field and exploit $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$.

The following code does just that to find all candidate $g'(x)$:



```
extendedDecompose[poly_, x_, extension_:Automatic] /; PolynomialQ[poly, x] :=
Module[{ext, res},
  ext = If[extension === Automatic, CoefficientList[poly, x], extension];
  res = FixedPoint[
    Flatten[Prepend[Rest[#], iExtendedDecompose[First[#], x, ext]]]&,
    {poly}
  ];
  res = res //. {a___, x^n_., x^m_., b___} := {a, x^(n + m), b};
```



```

    simplifyCoefficients[res, x]
]

extendedDecompose[expr_, ___] := {expr}

iExtendedDecompose[poly_, x_, extension_] :=
Module[{deg, facs, cands, dg, d, a, gcoeffs, f, g, sys, sol, res},
  deg = Exponent[poly, x];
  If[deg <= 1 || PrimeQ[deg], Return[{poly}]];

  facs = Select[FactorList[D[poly, x], Extension -> extension], Exponent[#
[[1]], x] > 0&];
  If[Length[facs] < 2, Return[{poly}]];

  cands = Select[Times @@@ Subsets[Join @@ ConstantArray @@@ facs], 1 <=
Exponent[#, x] <= 0.5deg&];

  SetAttributes[a, Listable];
  Catch[
    Do[
      dg = Exponent[gprime, x]+1;
      d = deg/dg;
      f = fromCoefficients[a[Range[0, d]], x];

      gcoeffs = Prepend[CoefficientList[gprime, x]/Range[dg], 0];
      gcoeffs /= SelectFirst[gcoeffs, Not @* PossibleZeroQ];
      g = fromCoefficients[gcoeffs, x];

      sys = Thread[CoefficientList[Expand[f /. x -> g] - poly, x] == 0];
      sol = Solve[sys, a[Range[0, d]]];

      If[ListQ[sol] && Length[sol] > 0,
        Throw[{f /. First[sol], g}]
      ],
      {gprime, cands}
    ];
  {poly}
]

SetAttributes[simplifyCoefficients, Listable];

simplifyCoefficients[poly_, x_] :=
  fromCoefficients[Simplify[CoefficientList[poly, x]], x]

fromCoefficients[coeffs_, x_] := coeffs . PowerRange[1, x^(Length[coeffs] - 1),
x]

```

Your example:

```

poly = 3 + 3  $\sqrt{2}$  + (14 + 4  $\sqrt{2}$ ) x + (12 + 26  $\sqrt{2}$ ) x^2 +
(56 + 8  $\sqrt{2}$ ) x^3 + (8 + 48  $\sqrt{2}$ ) x^4 + 48 x^5 + 16  $\sqrt{2}$  x^6;

decomp = extendedDecompose[poly, x]

```

```
{3 (1 + √2) + (14 + 4 √2) x + (4 + 12 √2) x^2 + 8 x^3, x + √2 x^2}
```

```
PossibleZeroQ[poly - (First[decomp] /. x -> Last[decomp]])]
```

```
True
```

A nested example:

```
poly2 = Collect[poly /. x -> x^4 - x + 1, x];
```

```
decomp2 = extendedDecompose[poly2, x]
```

```
{3 (47 + 35 √2) + (-478 - 368 √2) x + (540 + 436 √2) x^2 - 8 (25 + 22 √2) x^3,
 x - (2 x^2)/(4 + √2), x - x^4}
```

```
Fold[#1 /. x -> #2 &, decomp2] - poly2 // PossibleZeroQ
```

```
True
```

Share Edit Follow Flag

edited Sep 22, 2019 at 18:25

answered Sep 22, 2019 at 0:49



Greg Hurst

39.7k

1

103

149

-
- 3 ▲ This is also the method used by `Decompose` (due to Alagar and Trinh if I remember correctly). It was implemented well before the `Extension` option in `Factor` and we never really revisited it.
 – Daniel Lichtblau Sep 22, 2019 at 14:02
-